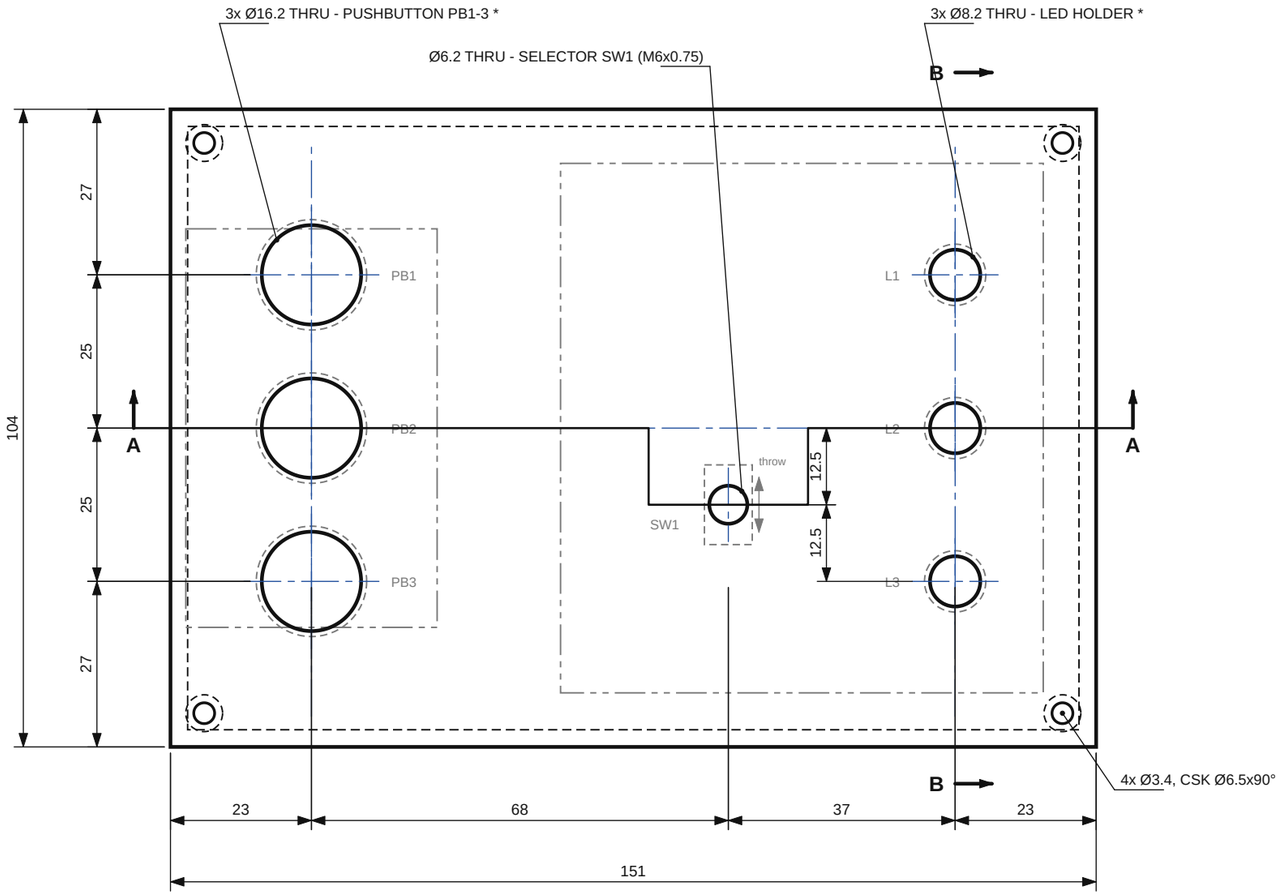
VIEW 1 - TOP (CONTROL PANEL / LID)

Lid 3.0 thick. Hidden: lid locating lip. Phantom: bays below.



VIEW 2 - PLAN, LID REMOVED

Battery bay LEFT under buttons | electronics bay RIGHT under LEDs

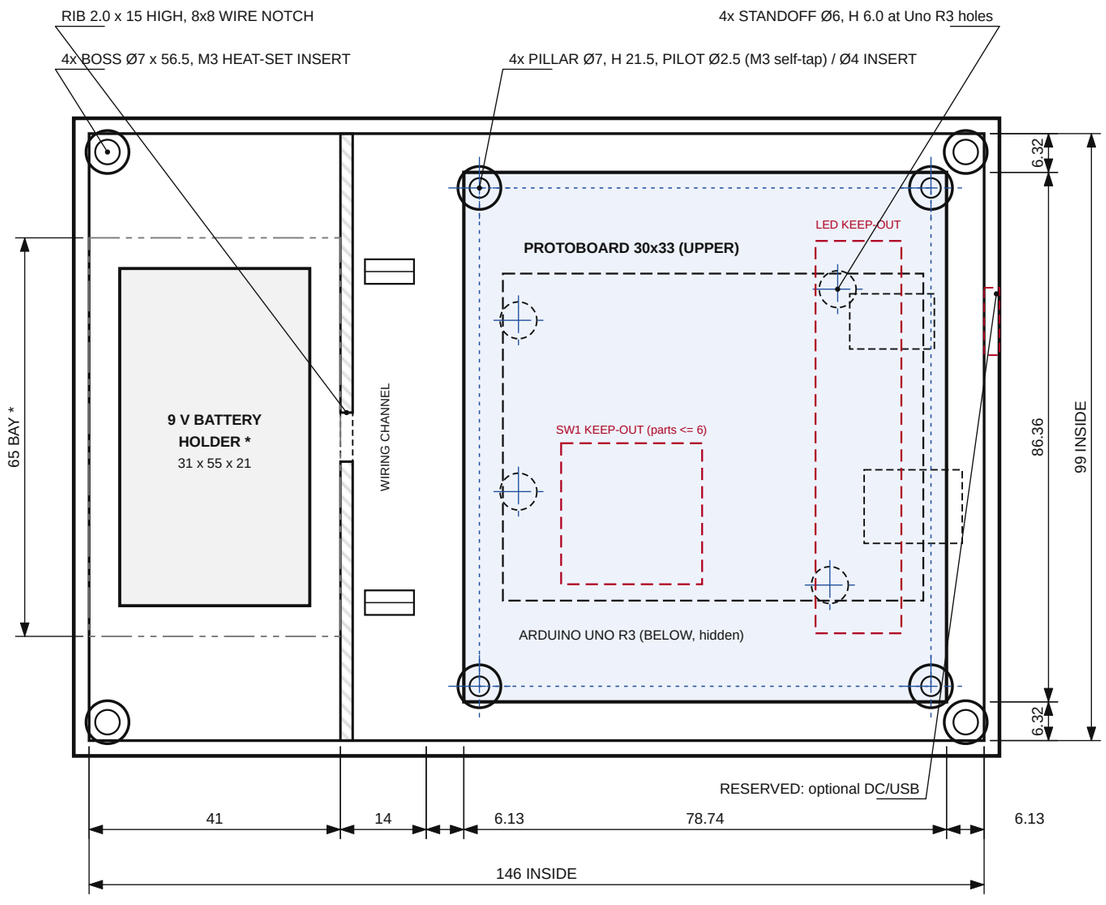


TABLE 1 - FEATURE COORDINATES (origin: external front-left-bottom corner, mm)

ITEM	X	Y	SIZE	NOTE
PB1/PB2/PB3	23	77 / 52 / 27	Ø16.2	Adafruit 1439, 16 mm illum. momentary *
SW1 selector	91	39.5	Ø6.2	Miyama MS-500E (ON)-OFF-(ON), M6x0.75
L1/L2/L3	128	77 / 52 / 27	Ø8.2	LED + 8 mm panel holder (model TBC) *
Lid screws	5.5 / 145.5	5.5 / 98.5	Ø3.4	4x M3 x 10 countersunk into inserts
Battery holder	7.5-38.5	24.5-79.5	31x55x21	on floor; VHB tape or 2x M3 *
Protoboard	63.63-142.37	8.82-95.18	78.74x86.36	underside Z 24.0 *
PB pillars	66.17 / 139.83	11.36 / 92.64	Ø7 x 21.5	corner holes drilled to Ø3.2
Arduino Uno R3	70-138.58	25.33-78.67	68.58x53.34	underside Z 8.5, jacks face +X
Uno standoffs	124.61 / 123.34 / 72.54 / 72.54	76.13 / 27.87 / 71.05 / 43.11	Ø6 x 6.0	4 holes, ref. Uno R3 drawing

** = nominal / assumed value - see Sheet 3, Table 4. Uno standoff X and Y are listed pairwise in the same order.*

LINE LEGEND

- Visible outline
- Hidden edge
- Centre line
- Zone / phantom (under lid)
- Keep-out / reserved

KEY DECISIONS (spec §1)

- LEFT: 9 V battery under button column
- RIGHT: protoboard above Uno R3, under LEDs
- SW1 right of centre, between L2/L3 rows (v6)
- CENTRE: protected wiring channel; no USB hole
- Envelope minimised: 151 x 104 x 62 (was 200x120x75)

REVISION HISTORY

A 2026-11-09 First issue from spec (9 Nov 2026).
B 2026-11-09 SW1 moved to layout-v6 position; PB/L numbered from back. * = assumed value.

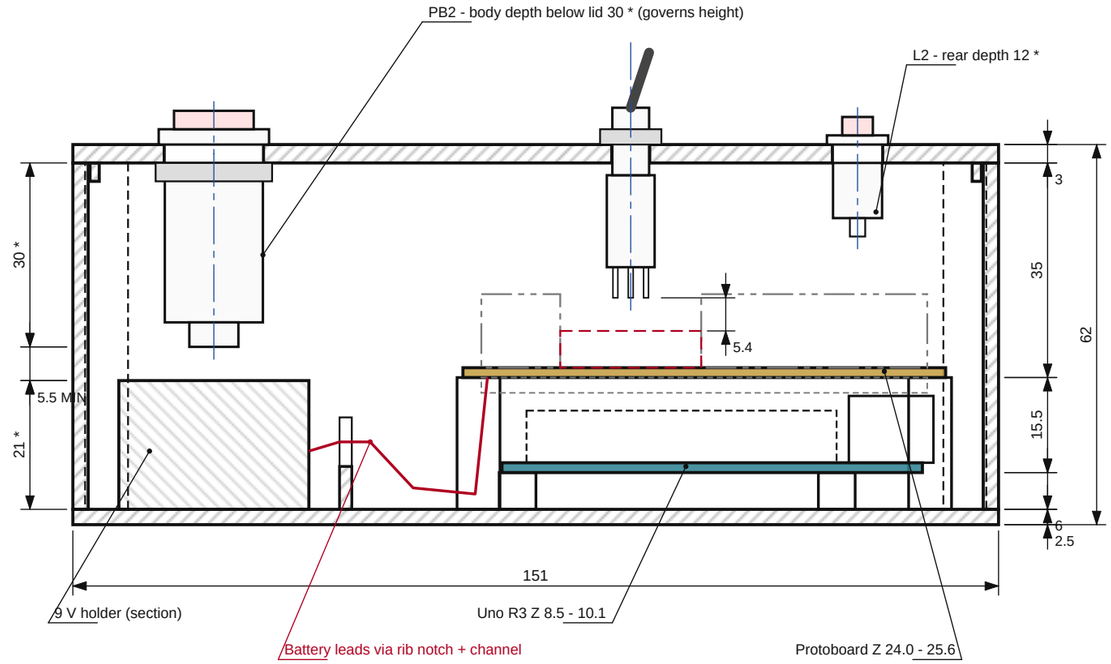
Drawn: Claude Code Checked: _____

Approved: _____

PROJECT: POWER GRID BOX - 3D PRINTED ENCLOSURE	DRAWING No.: PGB-ENC-001
TITLE: GENERAL ARRANGEMENT - PLAN VIEWS	SHEET: 1 OF 3
UNITS / SCALE: mm 1:1 @ A3	REV: B - PRELIM.
DATE: 2026-11-09	PROJECTION
MATERIAL / PROCESS: PETG, FDM, 0.2 mm layers	GENERAL TOL.: +/-0.3 unless stated
STATUS: DESIGN BRIEF - NOT FOR PRODUCTION	

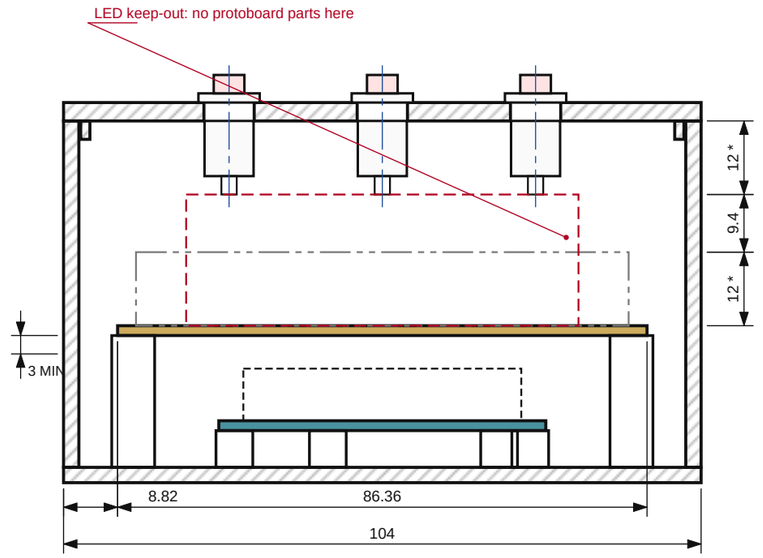
SECTION A-A (stepped, viewed from front)

Stepped section: Y = 52 (PB2, L2), jog to Y = 39.5 through SW1



SECTION B-B (X = 128, viewed from right)

Through LED column L1-L3 and the board stack



ISOMETRIC - EXPLODED LID (front-left view, not to scale)

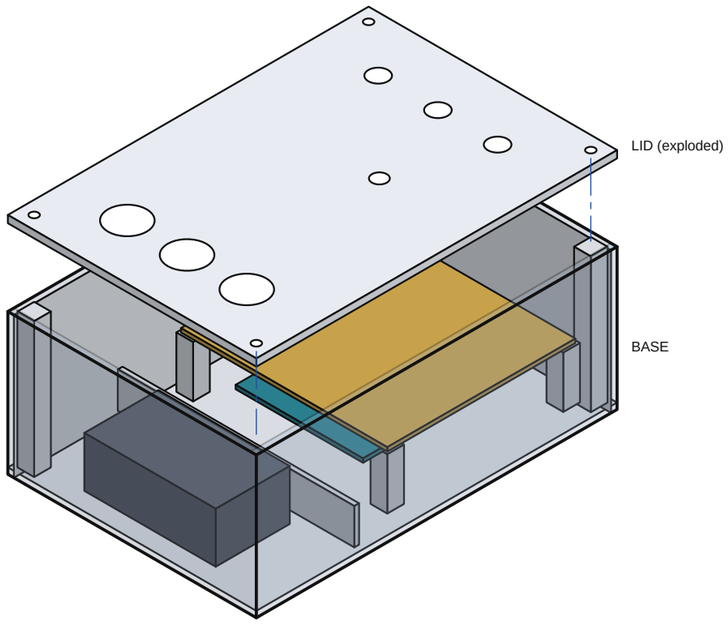


TABLE 2 - VERTICAL STACK-UP (Z from outside bottom)

LAYER	Z RANGE	NOTE
Floor	0 - 2.5	2.5 floor, 4x Ø10 x 1 foot recesses
Uno standoffs	2.5 - 8.5	6.0 high, Ø6
Uno R3 PCB	8.5 - 10.1	1.6 PCB
Uno parts / headers	10.1 - 21.0	DC jack/USB 10.9, headers 8.5
Clearance	21.0 - 24.0	3 (solder tails of protoboard)
Protoboard	24.0 - 25.6	on 4 floor pillars, H 21.5
Protoboard parts	25.6 - 37.6	max 12 * (keep-out under LEDs)
LED rear	47.0 - 59.0	12 *; 9.4 clear to parts
Battery holder	2.5 - 23.5	21 * high, on floor
Selector rear	37 - 59	22 *; parts under SW1 <= 6
Button rear	29.0 - 59.0	30 *; 5.5 clear to holder
Base top / lid	59.0 - 62.0	3.0 lid, 1.5x3 lip, 0.3 gap

Overall height is set by: holder 21 + 5.5 gap + button 30 + lid 3 + floor 2.5 = 62.

Each 1 mm saved on the measured button depth reduces height by 1 mm.

REVISION HISTORY

A 2026-11-09 First issue from spec (9 Nov 2026).
B 2026-11-09 SW1 moved to layout-v6 position; PB/L numbered from back. * = assumed value.

Drawn: Claude Code Checked: _____

Approved: _____

PROJECT: POWER GRID BOX - 3D PRINTED ENCLOSURE			DRAWING No.: PGB-ENC-001
TITLE: SECTIONS A-A / B-B, STACK-UP, ISOMETRIC			SHEET: 2 OF 3
UNITS / SCALE: mm 1:1 @ A3	REV: B - PRELIM.	DATE: 2026-11-09	PROJECTION
MATERIAL / PROCESS: PETG, FDM, 0.2 mm layers	GENERAL TOL.: +/-0.3 unless stated	STATUS: DESIGN BRIEF - NOT FOR PRODUCTION	

GENERAL NOTES

1. All dimensions in mm. Third-angle projection. Model origin: external front-left-bottom corner (X right, Y back, Z up).
2. Two printed parts: BASE (tub with floor, walls, rib, bosses, pillars, standoffs) and LID (control panel). Print lid face-down.
3. Walls 2.5, floor 2.5, lid 3.0. External corners R3 (vertical edges) optional; internal fillets R1 at wall/floor joints.
4. Lid fixing: 4x M3 x 10 countersunk screws into M3 x 5.7 heat-set inserts (Ø4.0 pilot, 6 deep) in corner bosses Ø7.
5. Lid lip 1.5 wide x 3.0 deep, 0.3 clearance per side to inner wall; notch lip 8 x 8 around each corner boss.
6. Panel holes are printed at +0.2 over nominal (Ø16.2 / Ø6.2 / Ø8.2). Ream to fit after a test print of a 30 x 30 coupon.
7. SW1 anti-rotation: add a Ø2.5 x 1.5 deep key hole at 8 mm from SW1 centre once the switch's tab washer is confirmed.
8. Protoboard on 4 floor pillars (Ø7 x 21.5) at its corner hole positions, drilled to Ø3.2; M3 self-tapping or insert.
9. Uno R3 on 4 standoffs Ø6 x 6.0 (Ø2.5 pilot for M3 self-tap / nylon M3). Jacks face the right wall (+X).
10. Battery service: remove 4 lid screws, fold lid over to the LEFT on a 150 mm wire service loop; holder is directly accessible. The protoboard / Uno stack is NOT disturbed (spec §6).
11. Wiring: battery snap leads exit the bay through the 8 x 8 rib notch (Y 48-56), run in the 12 mm channel, enter the electronics bay near the protoboard edge. 2x cable-tie anchors in channel at Y 25 and Y 79. Deburr all edges.
12. No USB opening (spec §1). Right wall zone at the Uno DC jack (Y 64-75, Z 10-22) is RESERVED - do not place ribs there, so an optional DC/USB opening can be cut later. Never connect an external supply in parallel with the 9 V battery.
13. Protoboard keep-outs: LED column (X 121-135, Y 20-84) parts <= 3; SW1 (X 79.5-102.5, Y 28-51) parts <= 6.
14. Add 4x Ø10 x 1.0 deep recesses on the underside at (12,12) (139,12) (12,92) (139,92) for rubber feet.
15. SW1 position follows the top layout of draft v6 (position only; no v6 dimensions used): right of centre, midway between the L2 and L3 rows. Mount the toggle so the lever throws along Y (back = towards L2, front = towards L3).

TABLE 3 - CLEARANCE CHECKS (computed from model)

CHECK	ACTUAL	MIN	RESULT
Button body -> battery holder (Z)	5.5	5	OK
LED rear -> protoboard parts (Z)	9.4	5	OK
Uno tallest part -> protoboard (Z)	3	2.5	OK
Protoboard edge -> right wall (X)	6.13	5	OK
Protoboard edge -> channel (X)	6.13	5	OK
Protoboard edge -> front/back wall (Y)	6.32	5	OK
Uno USB tip -> right wall (X)	3.57	2	OK
Selector rear -> protoboard parts (Z)	5.4	5	OK
Button bezel -> bezel (Y)	7	5	OK
Corner boss -> protoboard corner	1.06	0.5	OK

OPEN ITEMS (spec §1, §7)

- External DC barrel input: OPEN, not modelled (zone reserved).
- Confirm button quantity (3) and LED model / quantity (3).
- Confirm 2.8 x 0.5 QC receptacle fit on button blades.
- Verify SW1 terminal arrangement with continuity meter.

TABLE 4 - ASSUMPTIONS REGISTER (* values: the spec lists these as TBC/approx. - measure with calipers and update the parameters)

ITEM	VALUE USED	BASIS / SOURCE	ACTION
Protoboard outline	78.74 x 86.36 x 1.6	Hole span 73.66 x 81.28 (spec §4) + 2.54 edge margin each side	Measure edge-to-edge
Protoboard part height	12 max above board	Typical radial electrolytic / terminal block	Measure tallest part
Arduino Uno R3	68.58 x 53.34 x 1.6	Arduino reference design; holes (13.97,2.54) (15.24,50.80) (66.04,7.62) (66.04,35.56)	Verify on board
Uno part height	10.9 (USB-B / DC jack)	Reference design; headers 8.5	Verify
9 V battery holder	31 x 55 x 21	Spec §4 'approx. 55 x 31 x 21' - not verified	Measure + wire exit
Battery bay reserve	41 x 65	Spec §4 provisional planning reserve	Re-check after holder
Adafruit 1439 button	Ø16 mount, Ø18 bezel, 30 below panel	16 mm nominal (spec); depth incl. 2.8 mm QC terminal + wire bend ASSUMED	**CRITICAL: sets height**
Miyama MS-500E	Ø6.2 bore, body 7.8 x 13, 22 below panel	Spec §4 listing; depth below panel ASSUMED	Check mfr. drawing
LED indicator	Ø8.2 cut-out, 12 rear depth	Generic 5 mm LED in 8 mm panel holder - model not chosen	Select model
Panel thickness	3.0 lid	Within typical max panel thickness for M6 bushing / 16 mm button	Check both parts
Clearance allowance	>= 5 (6 at electronics bay)	Spec §6 preliminary allowance; 6 used so corner bosses clear the board	-

HOW TO UPDATE AFTER MEASURING

a) Edit the parameter block at the top of cad/power_grid_box_enclosure.scad (and tools/gen_drawing.py for this drawing set).
b) Overall size follows automatically: width = walls + battery bay + rib + channel + (2 x 6 + protoboard width);
depth = walls + 2 x 6 + protoboard depth; height = floor + holder + 5 + button depth + lid (re-check Table 3).
c) Re-run tools/gen_drawing.py - the clearance table (Table 3) and the asserts fail if any check drops below its minimum.
d) Before the full print: print the lid only (approx. 1.5 h) and trial-fit PB1-3, SW1 and L1-3.

REVISION HISTORY

A 2026-11-09 First issue from spec (9 Nov 2026).
B 2026-11-09 SW1 moved to layout-v6 position;
PB/L numbered from back. * = assumed value.

Drawn: Claude Code Checked: _____

Approved: _____

PROJECT: **POWER GRID BOX - 3D PRINTED ENCLOSURE**	DRAWING No.: **PGB-ENC-001**
TITLE: **NOTES, ASSUMPTIONS REGISTER, VERIFICATION**	SHEET: **3 OF 3**
UNITS / SCALE: mm 1:1 @ A3	REV: B - PRELIM.
DATE: 2026-11-09	PROJECTION
MATERIAL / PROCESS: PETG, FDM, 0.2 mm layers	GENERAL TOL.: +/-0.3 unless stated
STATUS: **DESIGN BRIEF - NOT FOR PRODUCTION**	